

13. Write a Pandas program to detect missing values of a given DataFrame. Display True or False.

Aim

To detect missing values in a DataFrame using Pandas.

Procedure

1. Import Pandas and NumPy.
2. Create a DataFrame with some missing values.
3. Use `isnull()` to detect missing values.
4. Display the result as True or False.
5. Verify the missing values.

Program and output

```
import pandas as pd
import numpy as np
# Create DataFrame
data = {
    "Name": ["John", "Alice", "David", "Emma"],
    "Age": [25, np.nan, 30, 28],
    "Salary": [50000, 60000, np.nan, 55000]
}
df = pd.DataFrame(data)
print("Original DataFrame:")
print(df)
print("\nMissing Values (True/False):")
print(df.isnull())
```

```

... Original DataFrame:
      Name  Age  Salary
0   John  25.0  50000.0
1  Alice   NaN  60000.0
2  David  30.0     NaN
3   Emma  28.0  55000.0

Missing Values (True/False):
      Name  Age  Salary
0  False  False  False
1  False   True  False
2  False  False   True
3  False  False  False

```

Result

Thus, the missing values in the DataFrame were detected successfully using `isnull()`.

14. Write a Pandas program to find and replace the missing values in a given DataFrame which do not have any valuable information.

Aim

To replace missing values in a DataFrame using Pandas.

Procedure

1. Import Pandas and NumPy.
2. Create a DataFrame with missing values.
3. Use `fillna()` to replace missing values.
4. Display the updated DataFrame.
5. Verify that missing values are replaced.

Program and output

```

import pandas as pd
import numpy as np

# Create DataFrame
data = {
    "Name": ["John", "Alice", "David", "Emma"],
    "Age": [25, np.nan, 30, 28],
    "Salary": [50000, 60000, np.nan, 55000]
}

```

```

df = pd.DataFrame(data
print("Original DataFrame:")
print(df)

# Replace missing values

df = df.fillna(0)

print("\nDataFrame after replacing missing values:")
print(df)

```

```

... Original DataFrame:
   Name  Age  Salary
0  John  25.0  50000.0
1  Alice  NaN  60000.0
2  David  30.0     NaN
3  Emma  28.0  55000.0

DataFrame after replacing missing values:
   Name  Age  Salary
0  John  25.0  50000.0
1  Alice   0.0  60000.0
2  David  30.0     0.0
3  Emma  28.0  55000.0

```

Result

Thus, the missing values in the DataFrame were successfully replaced using the fillna() function.